

## Lecture 5-DS

### Linear diff. equations in $\mathbb{R}^2$

#### 1. LHDE in $\mathbb{R}^2$ with CC

We consider a system with 2 equations and 2 unknowns.

$$(1) \begin{cases} x' = a_{11} \cdot x + a_{12} \cdot y \\ y' = a_{21} \cdot x + a_{22} \cdot y \end{cases}$$

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \in \mathcal{M}_2(\mathbb{R})$$

For  $X = \begin{pmatrix} x \\ y \end{pmatrix}$ :  $(1) \Leftrightarrow x' = A \cdot X$

- coupled, uncoupled sys.
- reduction method
- principal matrix
- exponential matrix

Def: A solution of (1) is a function  $\varphi: \mathbb{R} \rightarrow \mathbb{R}^2$ , of class  $C^1$  s.t.  
 $\varphi'(t) = A \cdot \varphi(t)$ ,  $(\forall) t \in \mathbb{R}$ .

We say that (1) is uncoupled when  $a_{12} = a_{21} = 0$ .  $\sim I$ .

$$\Rightarrow \begin{cases} x' = a_{11} \cdot x \\ y' = a_{22} \cdot y \end{cases} \quad (\text{dp.})$$

Otherwise, we say that (1) is coupled.

Proposition: The general sol. of an uncoupled system is  $\begin{cases} x = c_1 \cdot e^{a_{11} \cdot t} \\ y = c_2 \cdot e^{a_{22} \cdot t} \end{cases}$   
 $c_1, c_2 \in \mathbb{R}$ .

From now on, we consider that either  $a_{12} \neq 0$  or  $a_{21} \neq 0$ .

Remark: For a second order LHDE  $x'' + b_1 x' + b_2 x = 0$ ,  $b_1, b_2 \in \mathbb{R}$ ,  
we denote  $y = x'$  and we arrive to a LHDE in  $\mathbb{R}^2$ :

$$\begin{cases} x' = y \\ y' = -b_2 x - b_1 y \end{cases} \quad \left( \begin{aligned} y' = x'' &= -b_1 x' - b_2 x \\ &= -b_1 y - b_2 x \end{aligned} \right)$$

For this system, the coeff. matrix is  $A = \begin{pmatrix} 0 & 1 \\ -b_2 & -b_1 \end{pmatrix}$

The reduction method to find the general sol. of a coupled (1).

Ex. 1

Assume that  $a_{12} \neq 0$

Our aim: to find a relation between  $x''$ ,  $x'$  and  $x$  (to elim.  $y$ ).

$$x' = a_{11} \cdot x + a_{12} \cdot y$$

Step 1:  $y = \frac{1}{a_{12}} \cdot (x' - a_{11} \cdot x)$

Step 2:  $x'' = a_{11} \cdot x' + a_{12} \cdot y'$

Step 3: We use step 2 and the second eq.  $y' = a_{21} \cdot x + a_{22} \cdot y$ .

$$x'' = a_{11} \cdot x' + a_{12} \cdot (a_{21} \cdot x + a_{22} \cdot y)$$

Step 4: We use step 1 and step 3

$$x'' = a_{11} \cdot x' + a_{12} \cdot a_{21} \cdot x + a_{12} \cdot a_{22} \cdot \frac{1}{a_{12}} \cdot (x' - a_{11} \cdot x)$$

$$x'' - (a_{11} + a_{22}) \cdot x' + (a_{11} \cdot a_{22} - a_{12} \cdot a_{21}) \cdot x = 0 \quad (2).$$

Step 5: We find the gen. sol. of (2) :  $x = c_1 \cdot x_1 + c_2 \cdot x_2$

Step 6: We find  $y$  using step 1 and step 5.

Ex: Find the gen. sol. of  $\begin{cases} x' = -y \\ y' = 4x \end{cases}$

Step 1:  $y = -x'$

Step 2:  $\begin{cases} x'' = -y' \\ y' = 4x \end{cases} \Rightarrow x'' = -4x \Rightarrow x' + 4x = 0$

$$x^2 + 4 = 0$$

$$x^2 = -4$$

$$x_{1,2} = \pm 2i$$

$$\rightarrow \cos 2t, \sin 2t$$

$$x = c_1 \cdot \cos 2t + c_2 \cdot \sin 2t, \quad c_1, c_2 \in \mathbb{R}$$

$$y = -x' = -(-c_1 \cdot 2 \cdot \sin 2t + 2 \cdot c_2 \cdot \cos 2t)$$

The general sol:  $\begin{cases} x = c_1 \cdot \cos 2t + c_2 \cdot \sin 2t \\ y = -2c_2 \cdot \cos 2t + 2 \cdot c_1 \cdot \sin 2t \end{cases} \quad c_1, c_2 \in \mathbb{R}.$

Ex.1 Find the sol. of the IVP's

a)  $x' = -y$ ,  $y' = 4x$ ,  $x(0) = 1$ ,  $y(0) = 0$ .

$$\begin{pmatrix} x(0) \\ y(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

b)  $x' = -y$ ,  $y' = 4x$ ,  $x(0) = 0$ ,  $y(0) = 1$ .

$$\begin{pmatrix} x(0) \\ y(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

a) In the gen. sol (3), we plug  $t=0$ :  $x(0) = c_1$   
 $y(0) = -2c_2$

$$x(0) = 1; \quad y(0) = 0$$

$$\Rightarrow c_1 = 1 \quad \Rightarrow c_2 = 0$$

This IVP has a unique sol.  $\checkmark$  (3)

$$\varphi_a: \mathbb{R} \rightarrow \mathbb{R}^2, \quad \varphi_a(t) = \begin{pmatrix} \cos 2t \\ 2 \sin 2t \end{pmatrix}, t \in \mathbb{R}$$

b)  $c_1 = 0$ ;  $c_2 = -\frac{1}{2} \rightarrow$  unique values for the constants φ

$\Rightarrow$  This IVP has a unique sol.

$$\varphi_b: \mathbb{R} \rightarrow \mathbb{R}^2, \quad \varphi_b(t) = \begin{pmatrix} -\frac{1}{2} \sin 2t \\ \cos 2t \end{pmatrix}, t \in \mathbb{R}$$

Ex2: Consider  $E(t) = \begin{pmatrix} \cos 2t & -\frac{1}{2} \sin 2t \\ 2 \sin 2t & \cos 2t \end{pmatrix}$ ,  $A = \begin{pmatrix} 0 & -1 \\ 4 & 0 \end{pmatrix}$

Check that  $E'(t) = A \cdot E(t)$ ,  $t \in \mathbb{R}$  and  $E(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Find the eigenvalues of  $A$ .

Proof:

$$E(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \checkmark$$

$$E'(t) = \begin{pmatrix} -2 \cdot \sin 2t & -\cos 2t \\ 4 \cdot \cos 2t & -2 \sin 2t \end{pmatrix}$$

$$A \cdot E(t) = \begin{pmatrix} 0 & -1 \\ 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} \cos 2t & -\frac{1}{2} \sin 2t \\ 2 \sin 2t & \cos 2t \end{pmatrix} = \begin{pmatrix} -2 \sin 2t & -\cos 2t \\ 4 \cos 2t & -2 \sin 2t \end{pmatrix}$$

Eigenvalues:  $\boxed{\lambda^2 - (\text{Tr} A) \cdot \lambda + \det A = 0} \Rightarrow \lambda^2 + 4 = 0$

$$\text{Tr} A = 0 + 0 = 0; \quad \det A = -(-4) = 4.$$

$$\Rightarrow \lambda(\lambda + 4) = 0$$

$$\Rightarrow \lambda_{1,2} \in \{0, -4\}$$

$$\Rightarrow \lambda = \pm 2i$$

$$\Rightarrow \sigma(A) = \{-2i, 2i\}$$

Proposition: The characteristic eq. of (2) is:

$$\lambda^2 - (\text{Tr} A) \cdot \lambda + \det A = 0, \text{ where } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \text{ is}$$

the coeff. matrix of sys. (1).

$\Rightarrow$  The roots of the char. eq. of (2) are the eigenvalues of  $A$ .

Definition: The principal matrix solution of sys. (1):  $x' = AX$  is a  $C^1$  function  $E: \mathbb{R} \rightarrow M_2(\mathbb{R})$  s.t. 
$$\begin{cases} E'(t) = A \cdot E(t), t \in \mathbb{R} \\ E(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{cases}$$

Proposition: Let  $p_a: \mathbb{R} \rightarrow \mathbb{R}^2$  be the solution of the IVP  $x' = AX, x(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$  and let  $p_b: \mathbb{R} \rightarrow \mathbb{R}^2$  be the sol. of the IVP  $x' = AX, x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ . Then  $E: \mathbb{R}^2 \rightarrow M_2(\mathbb{R}), E(t) = (p_a(t), p_b(t)), t \in \mathbb{R}$  is the principal matrix sol. of  $x' = AX$ .

Proof:  $p_a'(t) = A \cdot p_a(t)$

$$p_a(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$p_b'(t) = A \cdot p_b(t)$$

$$p_b(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow E(0) = (p_a(0), p_b(0)) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \checkmark$$

$$E'(t) = (p_a'(t), p_b'(t)) = \begin{pmatrix} A \cdot p_a(t) & A \cdot p_b(t) \end{pmatrix} \Rightarrow$$

$$A \cdot E(t) = A \cdot (p_a(t), p_b(t))$$

$$= (A \cdot p_a(t), A \cdot p_b(t))$$

$$\Rightarrow E'(t) = A \cdot E(t), (\forall) t \in \mathbb{R}.$$

Theorem: For any  $\eta \in \mathbb{R}^2$ , the IVP  $x' = Ax$ ,  $x(0) = \eta$  has a unique sol. and this is  $\varphi(t, \eta) = E(t) \cdot \eta$ , where  $E(t)$  is the principal matrix sol. of  $x' = Ax$ .

Proof:  $\varphi(t) = E(t) \cdot \eta$ . We want to check that  $\begin{cases} \varphi'(t) = A \cdot \varphi(t), t \in \mathbb{R} \\ \varphi(0) = \eta. \end{cases}$

vector

$$\varphi(0) = E(0) \cdot \eta = I_2 \cdot \eta = \eta. \checkmark$$

$$\varphi'(t) = E'(t) \cdot \eta = A \cdot \underbrace{E(t)} \cdot \eta = A \cdot \underbrace{\varphi(t)} \checkmark$$

How to find directly  $E(t)$ ?

?  $E'(t) = a \cdot E(t)$ ,  $E(0) = 1$ ,  $a \in \mathbb{R}$

$$E(t) = e^{a \cdot t}, t \in \mathbb{R}.$$

$$\begin{cases} x' = ax \rightarrow x' - ax = 0 \\ x(0) = 1 \end{cases} \quad \begin{matrix} \kappa - a = 0 \\ \kappa = a \end{matrix}$$

$$x' = x \quad e^{at}$$

$$x(0) = 1$$

$$x = c \cdot e^{a \cdot t}$$

$$e^t$$

$$x(0) = c = 1$$

$$\text{if } c \cos t$$

$$\cos 0 = 1.$$

The matrix exponential

• For  $a \in \mathbb{R}$ , we have  $e^a = \sum_{k=0}^{\infty} \frac{1}{k!} \cdot a^k$

$$= 1 + a + \frac{1}{2!} \cdot a^2 + \frac{1}{3!} \cdot a^3 + \dots$$

↳ absolutely convergent

• For  $z \in \mathbb{C}$ , we have that  $e^z = \sum_{k=0}^{\infty} \frac{1}{k!} \cdot z^k$  abs. conv.

In addition, for any matrix  $A \in \mathcal{M}_2(\mathbb{R})$ , we have that

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k \text{ is abs. conv.}$$

How to compute  $e^A$ ?? for the moment, with Maple :

Theorem: For  $x' = Ax$ , the principal matrix sol. is  $E(t) = e^{tA}$ ,  $t \in \mathbb{R}$ .

Thus, for any  $\eta \in \mathbb{R}^2$  the solution of the IVP  $x' = Ax$ ,  $x(0) = \eta$  is

$$\varphi(t, \eta) = e^{tA} \cdot \eta.$$

Ex: For  $D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ , compute  $e^{tD}$  using the definition of the exponential matrix.

$$e^{tD} = \sum_{k=0}^{\infty} \frac{1}{k!} (tD)^k$$

$$= \sum_{k=0}^{\infty} \frac{t^k}{k!} \cdot D^k$$

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \Rightarrow D^2 = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} \lambda_1^2 & 0 \\ 0 & \lambda_2^2 \end{pmatrix}$$

diag. matrix diag. matrix.

Induction:  $D^k = \begin{pmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{pmatrix}, \forall k \in \mathbb{N}.$

$$\Rightarrow e^{tD} = \sum_{k=0}^{\infty} \begin{pmatrix} \frac{t^k \cdot \lambda_1^k}{k!} & 0 \\ 0 & \frac{t^k \cdot \lambda_2^k}{k!} \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{k=0}^{\infty} \frac{1}{k!} (\lambda_1 t)^k & 0 \\ 0 & \sum_{k=0}^{\infty} \frac{1}{k!} (\lambda_2 t)^k \end{pmatrix}$$

$$= \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix}$$

Lemma:  $\lambda_1, \lambda_2 \in \mathbb{R}.$

$$e^{t \cdot \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}} = \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix}, t \in \mathbb{R}.$$

Lemma:

Let  $A, B, P \in M_2(\mathbb{R})$ .

Assume  $P$  invertible and  $A = P \cdot B \cdot P^{-1}$

Then  $e^{tA} = P \cdot e^{tB} \cdot P^{-1}$ .